🚗 SalesRadar — Complete Study Guide

Interview se pehle ye zaroor padho!

PART 1: PROJECT OVERVIEW — KYA BANAYA?

SalesRadar kya hai? (Hinglish)

SalesRadar ek **full-stack analytics platform** hai jo car dealerships ke liye banaya gaya hai. Isme **Java (Spring Boot)** backend handle karta hai aur **Python (Streamlit)** frontend/ML karta hai. Dono technologies ko ek saath use karke ek powerful dashboard bana diya jisme:

- CSV upload hoti hai
- Data analyze hota hai
- AI se questions pooch sakte hain
- ML se future sales predict hota hai
- PDF report generate hoti hai

SalesRadar in English (Interviewer ko batane ke liye)

SalesRadar is a full-stack automotive sales analytics platform that bridges Java's enterprise capabilities with Python's data science ecosystem. It allows car dealerships to upload their sales data and get real-time insights through interactive visualizations, AI-powered natural language querying, and ML-based sales forecasting.

PART 2: TECHNOLOGIES USED — KYA KYA USE KIYA?

⚫ BACKEND TECHNOLOGIES (Java Side)

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1. Spring Boot 3

Hinglish: Spring Boot ek Java framework hai jo REST APIs banane ke liye use hota hai. Bina zyada configuration ke production-ready application ban jaata hai. Hamare project mein Spring Boot ne ye kaam kiya:

- CSV upload receive karna
- Database se data fetch karna
- Analytics endpoints banana
- PDF generate karna

English (Interview): Spring Boot 3 is the core backend framework used to build RESTful APIs. It follows layered architecture — Controller handles HTTP requests, Service contains business logic, and Repository manages database operations. Spring Boot's auto-configuration reduces boilerplate code significantly.

Jo padhna chahiye:

- Spring Boot Architecture (Controller-Service-Repository pattern)
- REST API concepts (GET, POST, DELETE)
- @RestController, @RequestMapping, @GetMapping, @PostMapping annotations
- Dependency Injection (@Autowired, Constructor injection)
- application.yml configuration

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2. Spring AI + Ollama (AI Integration)

Hinglish: Spring AI ek library hai jo Java applications mein AI/LLM integrate karne ke liye use hoti hai. Ollama ek tool hai jo LLM (Large Language Models) ko locally run karta hai — matlab internet ki zaroorat nahi! Hamare project mein user jo bhi question poochta hai (English/Hindi mein), wo Ollama ke through AI model ko jaata hai aur answer aata hai.

English (Interview): Spring AI provides seamless LLM integration within Spring Boot applications. We used Ollama to run the LLM locally, ensuring data privacy as no data leaves the system. The AI module allows users to query their sales data using natural language — for example, "Which state had highest sales in 2023?"

Jo padhna chahiye:

- What is LLM (Large Language Model)?
- What is Ollama? (Local AI runner)
- Spring AI basic concepts
- What is RAG (Retrieval Augmented Generation)? — mention kar sakte ho future scope mein

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3. Spring Data JPA + Hibernate

Hinglish: JPA (Java Persistence API) ek specification hai jo Java objects ko database tables se map karta hai. Hibernate iska implementation hai. Matlab — Java mein `CarSales` class likhi, aur automatically MySQL mein `car_sales` table ban gayi! SQL likhne ki zaroorat nahi — JPA methods se kaam hota hai.

English (Interview): Spring Data JPA with Hibernate serves as the ORM layer, mapping Java entities to MySQL tables. Custom JPQL queries were written for complex aggregations like yearly/monthly counts and brand/color/state wise analytics. The Repository pattern abstracts all database operations.

Jo padhna chahiye:

- ORM concept (Object Relational Mapping)
- JPA Annotations: @Entity, @Table, @Column, @Id, @GeneratedValue
- JPQL vs SQL difference
- @Query annotation for custom queries
- JpaRepository methods (findAll, save, deleteAll, count)

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4. MySQL

Hinglish: MySQL ek relational database hai jahan haara data store hota hai. Ek table hai `car_sales` jisme har car ki information store hoti hai — brand, color, price, state, year, etc.

English (Interview): MySQL serves as the primary relational database. The `car_sales` table stores all transactional data. We implemented data isolation by truncating the table before each new CSV upload, ensuring fresh analysis every time.

Jo padhna chahiye:

- Basic SQL (SELECT, INSERT, DELETE, GROUP BY, ORDER BY)
- Indexing basics
- Primary Key, Foreign Key concepts
- Aggregate functions (COUNT, AVG, MAX, MIN)

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5. iTextPDF 5

Hinglish: iTextPDF ek Java library hai jo programmatically PDF files generate karne ke liye use hoti hai. Hamare project mein jab user "Generate PDF" click karta hai, Spring Boot backend ek PDF banata hai jisme yearly summary, monthly data, aur KPI metrics hote hain — sab automatically!

English (Interview): iTextPDF 5 was used for automated report generation. The PDF includes KPI metrics, yearly sales summary table, monthly breakdown, and footer — all generated dynamically from live database data without any manual effort.

Jo padhna chahiye:

- iTextPDF basic concepts (Document, PdfWriter, Paragraph, PdfPTable)

- ByteArrayOutputStream — PDF bytes memory mein store karna
- ResponseEntity<byte[]> — PDF bytes HTTP response mein bhejana

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6. Apache Commons CSV

Hinglish: Ye library CSV files parse karne ke liye use hoti hai. Jab user CSV upload karta hai, ye library har row read karti hai aur Java objects mein convert karti hai — phir database mein save hota hai.

English (Interview): Apache Commons CSV handles CSV parsing during file upload. Each row is parsed, validated, and mapped to a CarSales entity object, which is then persisted to MySQL. Invalid rows are counted separately and reported back to the user via the upload response.

□ FRONTEND & DATA SCIENCE TECHNOLOGIES (Python Side)

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7. Streamlit

Hinglish: Streamlit ek Python library hai jo data science dashboards banane ke liye use hoti hai. Bina HTML/CSS/JS likhe sirf Python se interactive web app ban jaati hai! Hamare project mein Streamlit ne ye kaam kiya:

- Navigation sidebar
- File upload widget
- Charts display
- Metrics cards
- PDF download button

English (Interview): Streamlit serves as the frontend framework, enabling rapid development of interactive dashboards using pure Python. It communicates with Spring Boot via REST API calls using the requests library. Cache management was implemented using @st.cache_data to optimize API call frequency.

Jo padhna chahiye:

- Streamlit basic components (st.title, st.metric, st.dataframe, st.plotly_chart)
- st.cache_data — caching kya hota hai aur kyun use kiya
- st.file_uploader — file upload handling
- st.sidebar — navigation
- st.spinner, st.success, st.error, st.warning

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8. Plotly Express

Hinglish: Plotly ek Python visualization library hai jo interactive charts banati hai. Hamare project mein Line chart, Bar chart, aur Donut Pie chart use kiye. "Interactive" matlab user chart pe hover kar sakta hai, zoom kar sakta hai.

English (Interview): Plotly Express was chosen over matplotlib for its interactive capabilities. Charts include line charts for trend analysis, bar charts for comparisons, and donut pie charts for distribution visualization. All charts use plotly_dark template matching the dashboard's dark theme.

Jo padhna chahiye:

- px.line(), px.bar(), px.pie() basic usage
- update_layout() — chart customization
- template="plotly_dark"
- Difference between Plotly and Matplotlib

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9. scikit-learn (ML Model)

Hinglish: scikit-learn ek Python ML library hai. Hamare project mein **Linear Regression** use ki — jo ek supervised learning algorithm hai. Ye past yearly sales data se pattern seekhta hai aur future years predict karta hai. R^2 score model ki accuracy batata hai — 0 se 1 ke beech (1 = perfect).

English (Interview): scikit-learn's LinearRegression model was trained on historical yearly sales data. The model learns the trend (slope) of sales over years and extrapolates for future predictions. R^2 score indicates goodness of fit — our test data showed low R^2 due to random variance in synthetic data; real dealership data with growth trends would yield significantly higher accuracy.

Jo padhna chahiye:

- What is Machine Learning?
- Supervised vs Unsupervised Learning
- What is Linear Regression?
- What is R^2 score (coefficient of determination)?
- model.fit(), model.predict(), model.score() — basic usage
- numpy reshape(-1,1) — kyun karna padta hai

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10. Pandas + NumPy

Hinglish: Pandas data manipulation ke liye use hoti hai — API se aaya data DataFrame mein convert hota hai. NumPy numerical operations ke liye use hoti hai — ML model ke liye arrays banana.

English (Interview): Pandas DataFrames are used to structure API response data for Streamlit charts. NumPy arrays are used for reshaping feature matrices for scikit-learn input. np.maximum() ensures predicted values never go negative.

🏗 ARCHITECTURE — PROJECT KAISE KAAM KARTA HAI?

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User (Browser)
     ↓
Streamlit Dashboard (Python — localhost:8501)
     ↓ HTTP REST API calls (requests library)
Spring Boot Backend (Java — localhost:8080)
     ↓                    ↓
MySQL Database         Ollama LLM
(Data storage)         (AI queries)
     ↓
iTextPDF
(Report generation)
     ↑
scikit-learn (Python side — in-memory ML)
```

Flow explanation:

1. User CSV upload karta hai Streamlit pe
2. Streamlit Spring Boot ko POST request bhejti hai
3. Spring Boot CSV parse karta hai, MySQL mein save karta hai
4. Streamlit analytics page pe jaata hai
5. GET requests se data aata hai Spring Boot se
6. Streamlit Plotly charts banata hai
7. AI question poochha → Spring Boot → Ollama → Answer
8. PDF button click → Spring Boot PDF generate karta hai → Download

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🔑 KEY DESIGN DECISIONS — KYU YE CHOICES LI?

Kyun Spring Boot + Python dono use kiya?

- Spring Boot → Enterprise-grade REST APIs, database handling, PDF generation
- Python → Data science ecosystem (Plotly, scikit-learn) jo Java mein utna mature nahi
- Dono ko combine karke best of both worlds mila

Kyun data isolation implement ki?

- Pehle dono CSV ka data mix ho raha tha
- Solution: Naya CSV upload karne se pehle `DELETE /clear` endpoint call hoti hai

- Ye real-world problem tha jo humne solve kiya

Kyun Local LLM (Ollama) use kiya?

- Cloud LLMs (OpenAI) mein data bahar jaata hai — privacy concern
- Ollama se LLM locally run hota hai — data secure rehta hai
- Internet connection ki zaroorat nahi

Kyun Linear Regression use kiya ML ke liye?

- Simple, interpretable model
- Less data mein bhi kaam karta hai
- Explainable AI — interviewer ko samjha sakte ho
- Future scope: ARIMA, Prophet for better time-series forecasting

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PART 3: REAL PROBLEMS SOLVED

1. **Manual Reporting** → PDF auto-generate hoti hai
2. **Data Mixing** → Data isolation implement ki
3. **Complex Queries** → AI se natural language mein pooch sakte hain
4. **Future Planning** → ML se sales predict hoti hai
5. **Multi-dimensional Analysis** → Brand/Color/State/Fuel/Payment wise breakdown
6. **No Technical Skills Needed** → Simple CSV upload, baaki sab automatic